

Master's Degree in Quantum Computing
Higher Polytechnic School

Quantum Bayesian Inference for Atmospheric Biosignature and Technosig- nature Characterization in Exoplanetary Systems

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This Master's Thesis addresses the fundamental computational bottlenecks inherent to probabilistic astrobiological inference in the era of high-resolution exoplanetary characterization. With current and upcoming spectroscopic observatories—such as the James Webb Space Telescope (JWST) and the Habitable Worlds Observatory (HWO)—generating multivariate, noisy, and highly coupled atmospheric spectra, classical probabilistic graphical models (Bayesian Networks) face severe algorithmic ceilings. Specifically, exact inference methods (e.g., Variable Elimination, Junction Tree) suffer from exponential memory explosion bounded by $\mathcal{O}(N \cdot d^{w+1})$ due to high treewidth induced by dense photochemical reaction networks, while stochastic approximations (Markov Chain Monte Carlo - MCMC) exhibit an asymptotic convergence rate constrained to $\mathcal{O}(1/\sqrt{M})$ alongside severe variance divergence when evaluating ultra-rare events such as industrial technosignatures (CFCs, non-equilibrium anomalies).

To overcome these limitations, this thesis designs, formalizes, and evaluates a Quantum Bayesian Network (QBN) architecture executed via Quantum Amplitude Estimation (QAE). By mapping planetary conditional probability distributions into quantum superposition states in a 2^n -dimensional Hilbert space and implementing conditioned unitary rotation cascades, the model eliminates exponential tree compilation memory overhead. Furthermore, by exploiting Grover-based iteration and Inverse Quantum Fourier Transforms (IQFT), the quantum inference framework achieves a quadratic asymptotic speedup, achieving a strict convergence rate bounded by $\mathcal{O}(1/M)$. This work provides a rigorous bridge from theoretical quantum algorithms to practical astrobiological data pipelines.